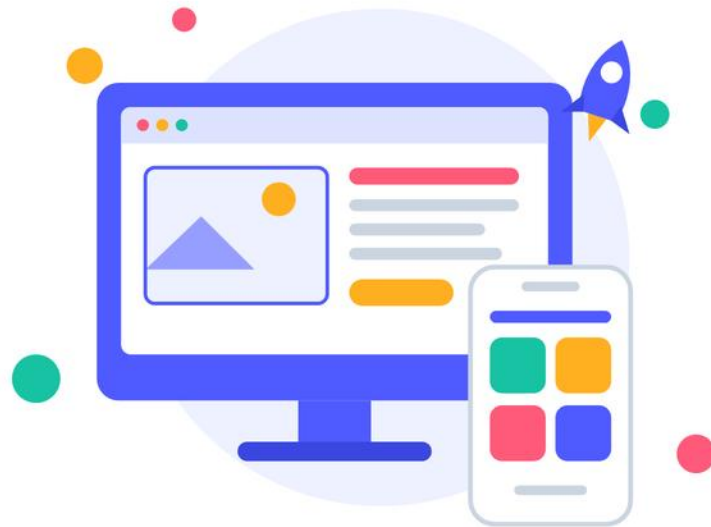


WEEK 04 · FLOW · POSITION · FLEXBOX

WHERE THE BOXES GO

Last week you learned to style a box. Today you learn to put one somewhere.

YOSSI ARVIV · FRONT-END WEB DEVELOPMENT · RUPPIN



WHERE WE ARE

Week 4 of 12. You have a page with a design language and no layout.

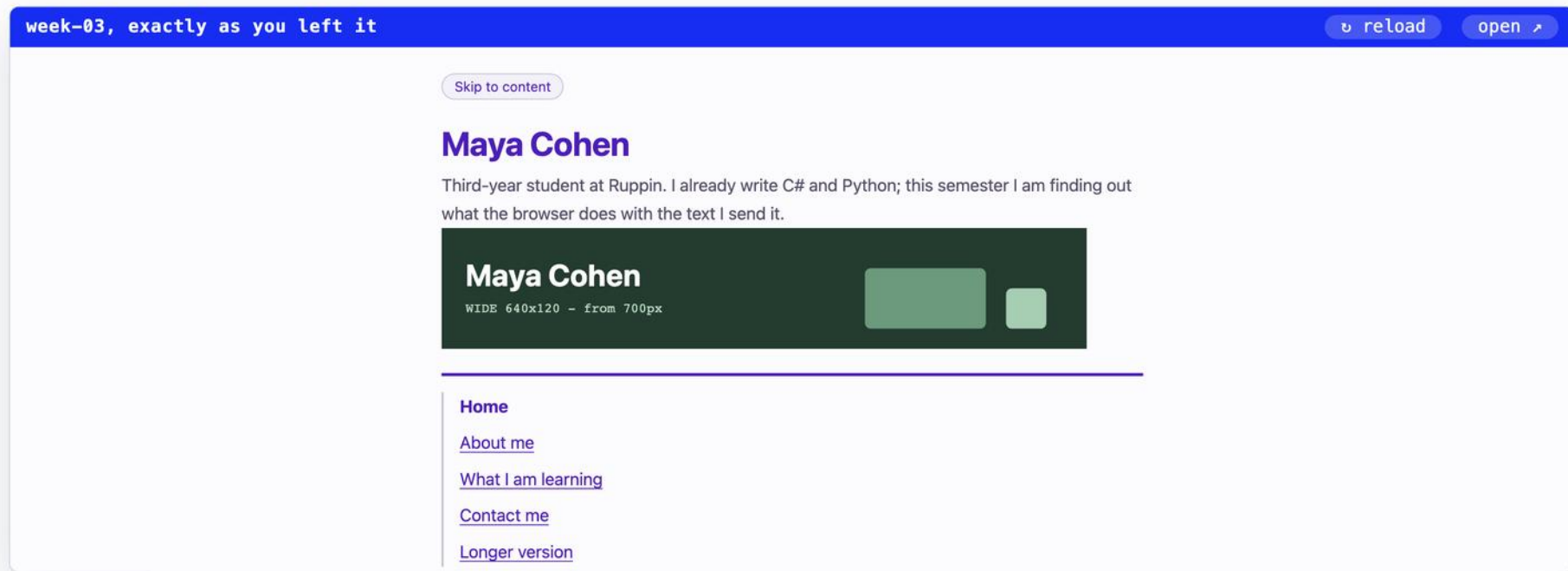
THE NEXT 180 MINUTES

Five cycles, you typing in every one: a navbar · a badge and a sticky header · a card row that wraps · one prompt to Claude · a menu that closes on a phone. Break at 90. The assignment at 145.

WHAT THIS UNLOCKS

Week 5 is Grid, and the decision rule "Grid for the page, Flex inside the component" only means something once you know what Flex is.

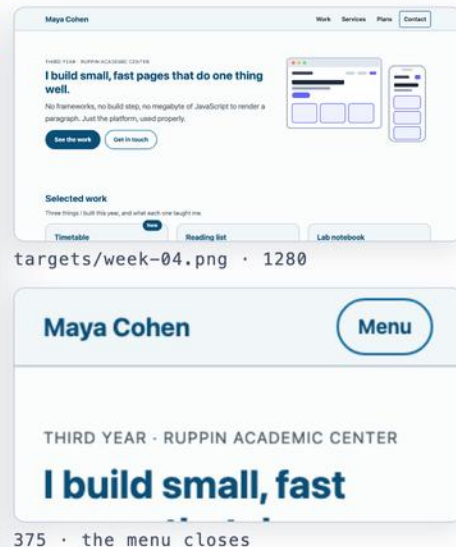
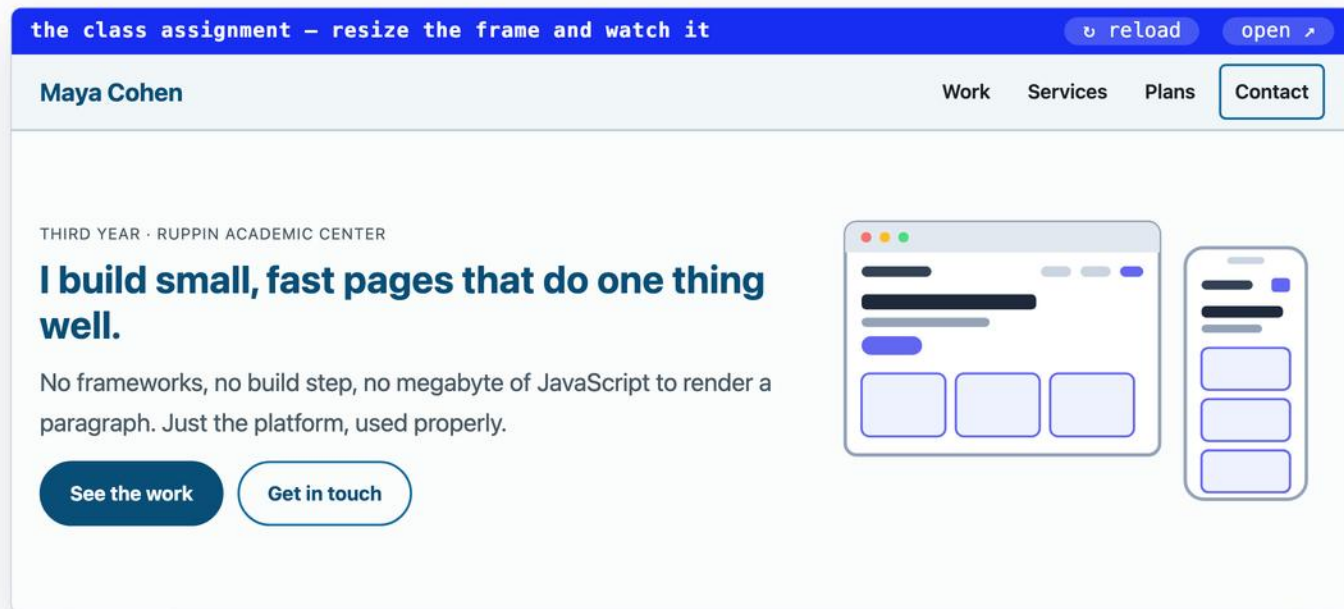
THIS IS EVERYTHING YOU CAN BUILD RIGHT NOW



► **SAY IT OUT LOUD** — this page is **GOOD**. And it is a **SINGLE COLUMN**, because that is the only shape you know how to make.

TARGET · MINUTE 0

THIS IS WHERE YOU WILL BE AT MINUTE 180



► **THE POINT** — everything that moves when you resize this does so without a single line that mentions a screen size. You type every piece of it today: once in a cycle, once in the assignment.

AI THIS WEEK

TUTOR. IT EXPLAINS; YOU TYPE.

LAYER · WEEKS 1-5 · PRIVATE TUTOR

ALLOWED

Ask it to explain the two axes, a stacking context, `flex-basis`. Ask it to criticise CSS *you* typed. Ask "why doesn't my rule apply?" — then check the answer in DevTools, not in your gut.

NOT ALLOWED

Pasting generated CSS into the assignment. **You type every character.** The tell this week: it writes Grid by default and media queries everywhere — both fail the assignment, and cycle 4 shows you why it does that.

NO SUBSCRIPTION?

Everything works in the free chat, and nothing in the course depends on a paid plan. Cycle 4 is one prompt, twice, in whatever chat you have.

CYCLE 1 · 25 MIN

FROM A STACK TO A NAVBAR

A link, then a list with bullets — exactly where normal flow put them. Six declarations later it is a navbar, and not one of them is a width.

1 I DO

10 MIN

3 on the next two slides, 7 in the editor · live-code/01-stack-to-row · from an empty bar to brand-left, links-right

2 WE DO

6 MIN

you type it on your copy · compare with checkpoint 1

3 YOU DO

7 MIN

warmup/ — the gold bar, five declarations, and the push that needs two

4 DEBRIEF

2 MIN

flex on the item · the ul padding · ↗ reference

YOU DO · AI-FREE

CYCLE 1 · BEFORE YOU TYPE

NOTHING YOU HAVE WRITTEN SO FAR HAS PLACED ANYTHING

Every page you have made was laid out by a rule you were never told. It is **two sentences long**.

A BLOCK BOX

Takes the full width offered; the next one starts underneath.

```
p h1 div section li
```

AN INLINE BOX

Takes the width of its content; the next one continues on the line.

```
a span strong em img
```

CYCLE 1 · BEFORE YOU TYPE

display DECIDES WHAT KIND OF BOX

It is not decoration and it is not visibility. It answers one question: *what sort of box does this element generate?*

block

Full width offered · stacks · width, height and all margins work.

inline

Content width · flows in a line · width, height and vertical margin ignored.

inline-block

Content flow, block sizing. Both, and one famous problem.

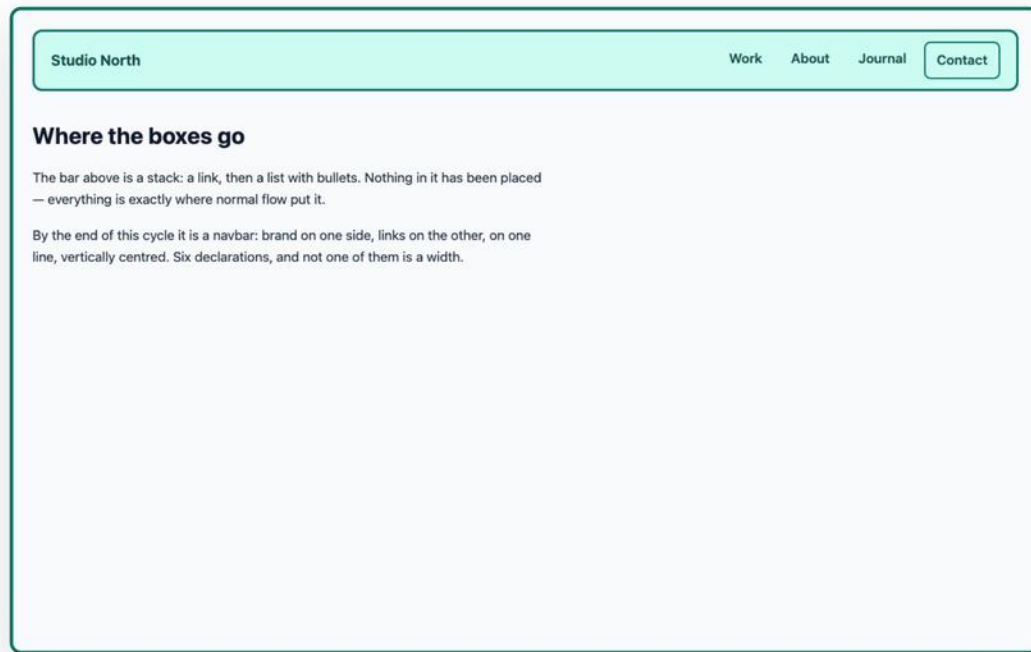
none

No box at all. Not "invisible" — absent.



CHECKPOINT 1

YOUR SCREEN SHOULD LOOK LIKE THIS



Compare — `live-code/01-stack-to-row/checkpoint.png` is this picture at 1280; `live-code/01-stack-to-row/end/` is the code behind it. Brand left, links pushed right, one line, Contact outlined.

CYCLE 2 · 25 MIN

THE BADGE, THE ANCHOR, AND THE HEADER THAT STAYS

Three uses of `position`, and not one of them places a column: a badge on a corner, the box it is measured from, a header that stays — above the content.

1 I DO

9 MIN

2 on the next slide, 7 in the editor ·
live-code/02-badge-and-anchor ·
the badge flies, the anchor, sticky,
the 999 that loses

2 WE DO

6 MIN

type it · checkpoint 2 · `offsetParent`
in the console

3 YOU DO

8 MIN

a second badge, bottom-left of the
third tile · verify with `offsetParent`

4 DEBRIEF

2 MIN

the diagnostic sentence · ↗
reference

YOU DO · AI AS TUTOR

EVERY BOX IS MEASURED AGAINST ANOTHER BOX

containing block The box your box's sizes and offsets are resolved against.

in normal flow Your nearest block-level ancestor's **content box**.

why you care `width: 50%` is 50% *of that*. So is `left: 10%`.

Change which box that is, and every percentage in the element changes meaning without you touching it. **Cycle 2 is entirely about changing it.**

```
1 .parent {  
2   width: 600px;    /* the containing block */  
3 }  
4  
5 .child {  
6   width: 50%;      /* 300px */  
7 }
```

CYCLE 2 · BEFORE YOU TYPE

TAKING A BOX OUT OF THE CONVERSATION

Normal flow is a negotiation between siblings. `position` is how one box *leaves* it — and the cost is that nothing reserves its space any more.

static

The default. In flow. Offsets ignored.

relative

In flow, keeps its space, moves visually.

Anchors its children.

absolute

Out of flow. Measured from the nearest positioned ancestor.

fixed

Out of flow. Measured from the viewport. Does not scroll.

sticky

In flow, then pins at a threshold. The one you will ship.



ALL FIVE, ON ONE SCROLLING PAGE

[demos/03-position/02-five-values](#)[↶ reload](#)[open ↷](#)

[static](#)[relative](#)[absolute](#)[fixed](#)[sticky](#)

THE DASHED FRAME IS THE CONTAINER. IT IS `position: relative`.

Text above the box.

where it would be in normal flow

the box

`position: static`

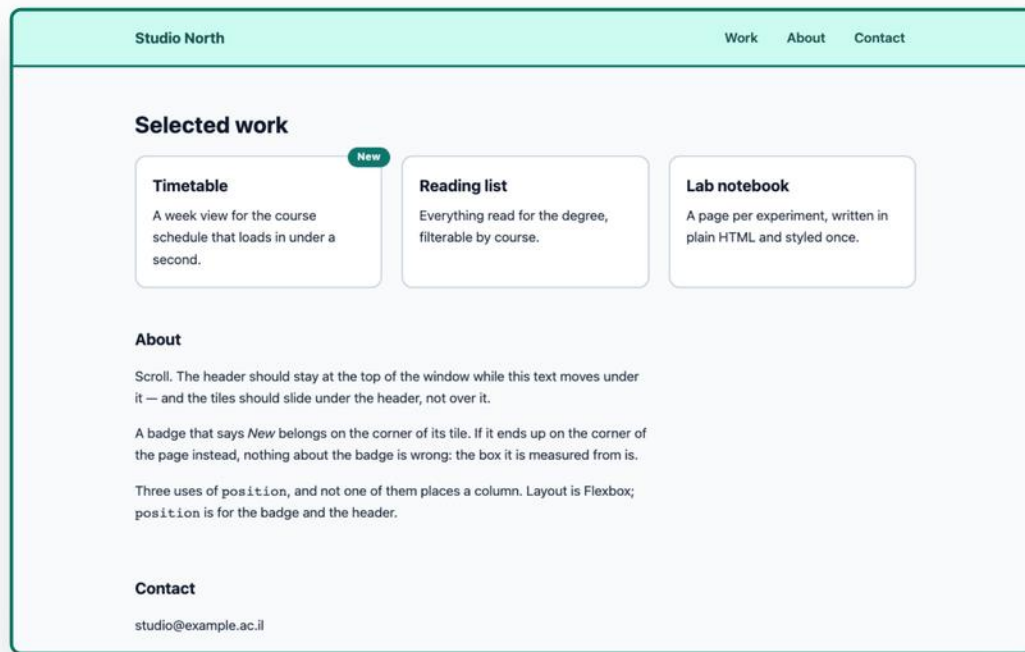
still in normal flow?	yes – this IS normal flow
offsets measured from	nothing, top and left are ignored entirely.
when you scroll	scrolls with the page
container height now	0px
computed position	static

The default. Every element on every page you have written so far.
Note that top/left do nothing here – offsets need a position first.

► **HANDS-ON** [demos/03-position/02-five-values/](#) — Start on `relative` and point at the dashed ghost: **THAT SPACE IS STILL RESERVED.**

CHECKPOINT 2

YOUR SCREEN SHOULD LOOK LIKE THIS



Compare — `live-code/02-badge-and-anchor/checkpoint.png` is this picture at 1280; `live-code/02-badge-and-anchor/end/` is the code behind it. The badge on the corner of the first tile, not of the page; the header stays when you scroll and the tiles slide under it.

KEEP THIS ONE. IT IS A DIAGNOSTIC, NOT A DEFINITION.

If it aligns to the page instead of the box you expected, you probably forgot `position: relative` on the ancestor.

FROM THIS COURSE, BEFORE YOU WERE IN IT

AND ITS TWO COMPANIONS

Absolute elements won't stretch the parent's height
— parents may collapse.

fixed inside a transformed ancestor behaves like
`absolute`. Avoid transforming high-level wrappers.

A STACKING CONTEXT IS A SEALED BOX

When an element creates one, its children's `z-index` values are only ever compared **with each other**. Against anything outside, what competes is *the parent's own* `z-index`.

So `999` was never being compared with `1`. It was being compared with its siblings, inside a box whose own `z-index` is `auto` — and `auto` loses to `1`.

Raising the number cannot help. It is the wrong argument.

position

...that is not `static`, plus any `z-index`

opacity

anything below `1`

transform

anything other than `none`

filter

anything other than `none`

will-change

naming any of the above

isolation

`isolate` — the only one that is deliberate

CYCLE 3 · 30 MIN

THE CARD ROW THAT WRAPS BY ITSELF

Three across when there is room, one under the other when there is not, nothing broken in between — and not one line that mentions a screen.

1 I DO

10 MIN

3 on the next three slides, 7 in the editor · live-code/03-card-row-wraps · width fails, flex-basis wins, min-width: 0, the link at the bottom

2 WE DO

8 MIN

type it · checkpoint 3 · a fourth card with no CSS change

3 YOU DO

10 MIN

a pricing row: three tiers, buttons on one line at the bottom

4 DEBRIEF

2 MIN

width vs flex-basis · min-width: auto · ↗ reference

YOU DO · AI-FREE

CYCLE 3 · BEFORE YOU TYPE

ONE DIMENSION, DONE PROPERLY

It's designed for components — navbars, cards, toolbars, forms — rather than full page grids.

FROM THIS COURSE, BEFORE YOU WERE IN IT

A **row or a **column**, with real control over how the space between and around the items is distributed. That is the entire scope, and it is why it is easy.**



TWO AXES, AND THEY MOVE

main axis The direction items are laid out. Set by `flex-direction`.

cross axis Perpendicular to it. Always.

justify-content Distributes along the **main** axis.

align-items Aligns along the **cross** axis.

Neither of those two is "horizontal" or "vertical". They are **main** and **cross**, and which is which is a decision you made one line earlier.

Where the axes point

FLEX-DIRECTION	MAIN	CROSS
<code>row</code>	horizontal →	vertical ↓
<code>row-reverse</code>	horizontal ←	vertical ↓
<code>column</code>	vertical ↓	horizontal →
<code>column-reverse</code>	vertical ↑	horizontal →

CONTAINER PROPERTIES · ITEM PROPERTIES

ON THE CONTAINER

`flex-direction` · `flex-wrap` · `justify-content` ·
`align-items` · `align-content` · `gap`

Controls the flow and the alignment of all the items at once.

ON AN ITEM

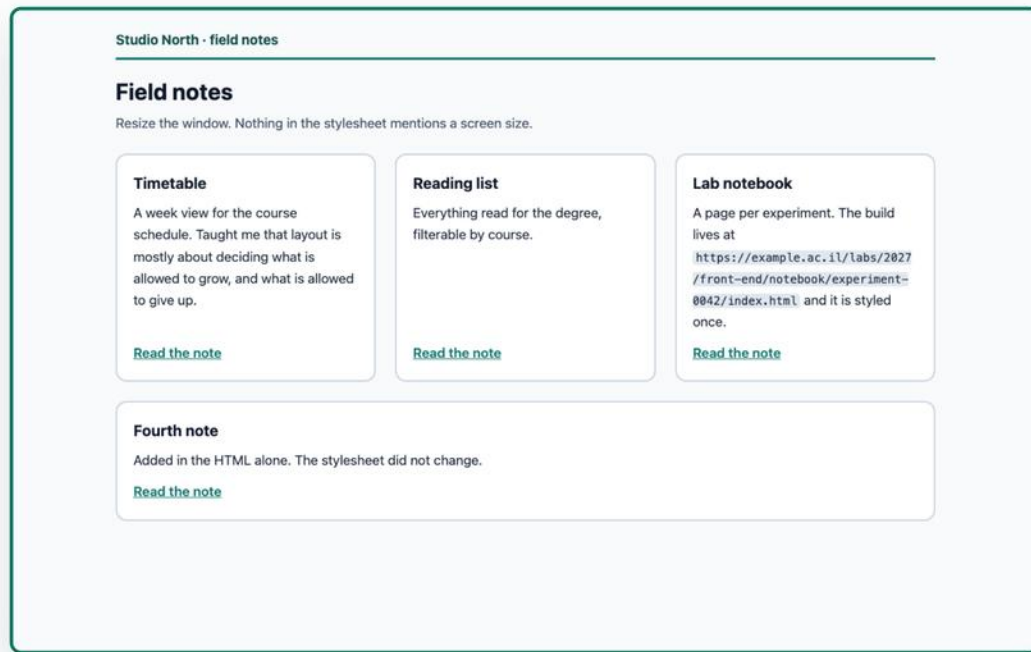
`flex-grow` · `flex-shrink` · `flex-basis` (and the
`flex` shorthand) · `align-self` · `order`

Controls the size and the alignment of one item.

Start with `flex-basis` to set the natural size,
then tweak `flex-grow` and `flex-shrink`.

CHECKPOINT 3

YOUR SCREEN SHOULD LOOK LIKE THIS



Compare — `live-code/03-card-row-wraps/checkpoint.png` is this picture at 1280; `live-code/03-card-row-wraps/end/` is the code behind it. Three across, equal height, links on one line, the URL broken inside its card, the fourth card alone and full width below.

BREAK

10 MINUTES

Back at 100. Cycle 4 is the AI cycle — have a chat window open, free or paid.

CYCLE 4 · 25 MIN

ONE PROMPT, TWICE: "PUT THESE THREE CARDS IN A ROW"

Paste a stylesheet that has one float in it and ask for a row. What comes back is 2010 — correct at 1280, wrong at the width you did not check. Then ask again with constraints.

1 I DO

7 MIN

2 on the next slide, 5 in the chat ·
ai-lab/04-floats-or-flex · P1 live · run
it at 1280, then at 375 ·
transcript.md if the room got
something else

2 WE DO

6 MIN

everyone pastes P1 · find what is
wrong before it is named

3 YOU DO

10 MIN

write P2 yourself · run it at three
widths · a journal line: claim ·
checked · verdict

4 DEBRIEF

2 MIN

the claim that has to be checked
between X and Y · ↗ exemplars

YOU DO · AI AS TUTOR

CYCLE 4 · BEFORE YOU PROMPT

TWO MINUTES ON `float`, BECAUSE YOU WILL MEET IT

demos/07-archaeology — five ways, 1998 to 2017

↻ reload

open ↗

1 · tables (1998)

2 · inline-block (2007)

3 · absolute (2005)

4 · float (2010)

5 · flex (2017)

THE SAME THREE COLUMNS. THE MARKUP NEVER CHANGES.

Column one, with a sentence in it so the boxes have something to be.

Column two.

Column three, which is the longest of the three by some way.

A paragraph after the columns.

```
.cols { display: table; width: 100%; border-spacing: 5px }  
.col { display: table-cell; width: 33.33% }
```

What it cost: the layout lives in the markup. A screen reader announces a table with rows and cells to someone who is trying to read an article, and the page cannot be rearranged for a narrow screen at all – a table is a table at every width.

Tables are for tabular data. That has never changed.

► **HANDS-ON** `demos/07-archaeology/` — Click through in order and read the "what it cost" line. Then stop.

CHECKPOINT 4

THREE QUESTIONS BEFORE THE NEXT CYCLE

1. Did you open the answer to P1 **at 375** and see the three narrow columns yourself — not just at 1280?
2. Does your P2 contain at least two conditions a browser can check — "one column at 375", "no media query" — and does the answer meet both when you check?
3. Does your journal line contain a **width or a measurement** — a number, a tool, what you typed in the console — and not just "worked"?

Compare — `ai-lab/04-floats-or-flex/checkpoint-he.md` has the same three questions; `artifact/claude-p1/` and `artifact/claude-p2/` are what came back, ready to open.

CYCLE 5 · 20 MIN

THE MENU THAT CLOSES ON A PHONE

Everything else on the page adapts with no media query. This is the one question Flexbox cannot answer — so this is the one query you are allowed.

1 I DO

7 MIN

2 on the next slide, 5 in the editor · live-code/05-menu-closes-on-mobile · the rules outside the query, then inside · the two given lines · the overlay

2 WE DO

5 MIN

type it · checkpoint 5 · check both widths

3 YOU DO

6 MIN

Contact first in the open panel with order · dividers · both undone above 48em

4 DEBRIEF

2 MIN

how many queries are in your file? · ↗ reference

YOU DO · AI AS TUTOR

RESPONSIVE, WITH NOTHING THAT MENTIONS A SCREEN — AND THE ONE THING THAT DOES

THE WHOLE CARD ROW

```
1 .card-row {
2   display: flex;
3   flex-wrap: wrap;
4   gap: 1.5rem;
5 }
6
7 .card {
8   flex: 1 1 17rem; /* I want 17rem.
9                     If you cannot,
10                     take a line. */
11 }
```

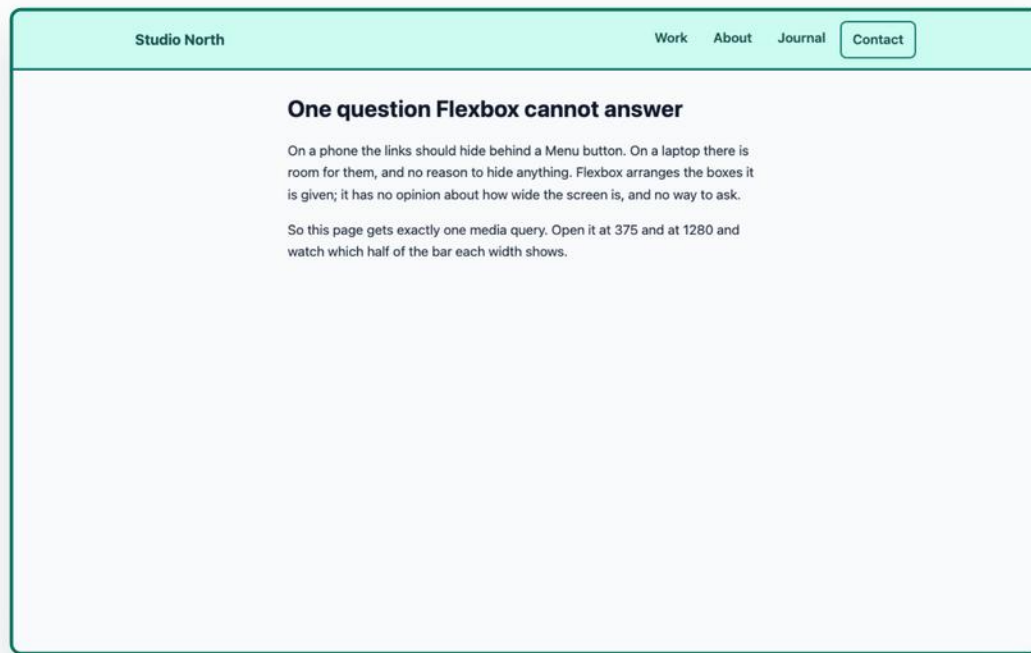
Three across at 1280. Two at 900. One at 375. **No media query, no breakpoint, no arithmetic that has to add up to 100 by hand.**

`flex-wrap` says items may take a new line rather than overflow; `flex-basis` says how much room each one wants before it gives up.

The one thing they cannot do: hide a Menu button on a laptop and show it on a phone. That is a question about the *viewport*, and it gets the one `@media (min-width: 48em)` in the whole file.

CHECKPOINT 5

YOUR SCREEN SHOULD LOOK LIKE THIS



Compare — `live-code/05-menu-closes-on-mobile/checkpoint.png` is this picture at 1280; `live-code/05-menu-closes-on-mobile/end/` is the code behind it. At 1280: links in a row, no button. Now 375 yourself: a Menu button, and the panel floats over the heading when opened — the heading does not move.

ASSIGNMENT · 145–175

THE ASSIGNMENT: ASSEMBLE WHAT YOU BUILT, THEN EXTEND IT

ASSEMBLE · CORE

`week-04/` is already in your repo from `pre-class/`. The navbar is cycle 1 + cycle 5. The badge and the sticky header are cycle 2. The card row is cycle 3. The hero is cycle 3 with two items instead of three.

EXTEND · STRETCH AND CHALLENGE

The pricing row is cycle 3's "you do" on the assignment's markup. The badge, the sticky header and the footer row are one declaration each — you typed all of them today.

THE RULES, AND THE EXAM

Flexbox only. No Grid, no float, one media query. AI as tutor — it explains, you type. Submit the SHA in Moodle. Week 13 asks you about this code, offline.

DEBRIEF · 175–180

THREE QUESTIONS BEFORE YOU LEAVE

1. A badge with `position: absolute; top: 0; right: 0` lands in the corner of the page instead of the corner of its card. What is missing, and on which element?
2. A dropdown has `z-index: 999`. A header has `z-index: 1`. The dropdown is behind. Give me one reason — and tell me why raising 999 will not help.
3. `flex-direction: column`. Which property now centres the items left-to-right, and which one distributes them top-to-bottom?

► **GO DEEPER** — the reference deck is optional depth, and it is not examined: every value of `position`, everything that creates a stacking context, every Flexbox property in two tables, and the demos behind today's cycles. The next slide lists it.

GO DEEPER

The reference deck for this week — every slide the lecture pointed at, in order.

[1 · The layout catalogue](#) ← spine 29

[2 · display — what kind of box](#) ← spine 5

[8 · Four ways to hide, and what each one costs](#) ← spine 25

[10 · position — the five values](#) ← spine 9

[14 · What each box is measured against](#) ← spine 9

[17 · What creates a stacking context](#) ← spine 9

[20 · overflow](#) ← spine 9

[23 · Flexbox · on the container](#) ← spine 5

[26 · Which axis is which](#) ← spine 16

[28 · Flexbox · on an item](#) ← spine 16

[31 · The flex shorthand](#) ← spine 16

[35 · @media \(min-width: 48em\) — the one you are allowed](#) ← spine 25

[36 · The five that will cost you time](#) ← spine 16

[37 · What float is still for](#) ← spine 22

[38 · Where to look things up](#) ← spine 29